Gold Money

Do Gold Prices Really Move Currencies?

Evidence from 21 years of daily data

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Coverage: AUD, ZAR, CAD, CHF versus gold, measured in Japanese yen
Sample: Daily, 4 January 2005 to 25 February 2026 (5,146 observations)
Asset classes: Gold (XAU), G10 and EM foreign exchange

Key Takeaways

- **The Australian dollar is the only genuine gold currency in this study, and only since 2020.** Gold now explains 63% of AUD's movement. The link holds even after removing the influence of the US dollar and market fear, which means it comes from real gold exports, not from a statistical accident.
- **The South African rand no longer deserves its gold-currency reputation.** Once prices are measured against a neutral currency and the dollar is accounted for, the rand's gold link disappears. The rand follows the dollar and global risk appetite, not gold.
- **Most gold-currency analysis is distorted by a simple measurement problem.** Gold is priced in US dollars, so charts drawn in dollars exaggerate the gold-currency relationship or even flip its sign. Measured in yen instead, the picture changes completely.
- **Nothing ties any currency to gold in the long run.** The statistical tests found no anchor pulling gold and these currencies back together. When they drift apart, they can stay apart for years.
- **Who moves first tells you what a currency really is.** Gold moves first for the mining currencies (AUD, CAD). The risk currencies (ZAR, CHF) move first for gold. This simple test can classify any candidate currency.
- **The relationship is barely tradable.** An 11-year backtest of a systematic strategy earned 4.5% on AUD before costs and lost money on the other three currencies. The link is real, but it switches off too often for fixed trading rules.

Summary assessment

Currency	Gold link (full sample)	Gold link (2020-26)	Survives controls?	Classification
AUD	Weak (R^2 0.11)	Strong (R^2 0.63)	Yes (b 0.16, t 12.2)	Genuine gold currency, regime-dependent
ZAR	Wrong sign (b -0.37)	Positive, dollar-driven	No (b -0.15)	Dollar-beta / risk currency
CAD	Weak (R^2 0.09)	Strong (R^2 0.60)	Yes (b 0.11, t 5.2)	Minor gold currency
CHF	Strongest (R^2 0.78)	Strongest (R^2 0.86)	Yes (b 0.31, t 24.1)	Safe-haven twin, zero mining exposure

The CHF row is the cautionary tale: the highest gold correlation in the sample belongs to a country with no gold mines. Correlation is not geology.

1. Investment Thesis

You have probably heard the claim. Gold is up, so the Australian dollar should be up too, because Australia digs a lot of gold out of the ground. The same is said about the South African rand. It sounds sensible, and it gets repeated far more often than it gets tested. This report tests it.

Gold is a tricky asset to test, because it is not just a commodity. Copper is something you use; gold is mostly something you hold. People buy it when they are frightened, when interest rates are low, or when they distrust the dollar. Those same forces move every currency on earth. So gold and the Australian dollar can rise together without either one causing the other. A naive correlation actually mixes up three different channels:

The income channel. When gold gets more expensive, a gold-exporting country earns more for the same shipments, so its currency should strengthen. This is the channel the market story assumes.

The denomination channel. Gold is priced in dollars. When the dollar weakens, the dollar price of gold rises and most currencies rise against the dollar at the same time, mechanically.

The common-factor channel. Fear and interest rates push gold and currencies around together without any real link between them. In a crisis, gold rises while the rand falls, and neither is causing the other.

1.1 Why these four currencies

The four currencies were chosen so that each one tests a different part of the story:

Currency	Role	Rationale
AUD	Primary candidate	Australia is a top-three gold producer, gold is among its biggest exports, and the currency floats freely. If any currency should follow gold, it is this one.
ZAR	Historic candidate	South Africa was the world's largest producer for most of a century, but its mines have been shrinking for twenty years. It tests whether the label survives when the industry fades.
CAD	Control case	Canada mines plenty of gold but is a big, diversified exporter (oil, autos, metals). If CAD shows the same gold link as AUD, the link is probably global, not mining-related.
CHF	Trap detector	Switzerland mines no gold at all, but the franc and gold are both safe havens. A strong gold correlation here proves that correlation can exist with zero gold exports.

Why not the other big producers? China consumes most of its own gold and manages its currency. Russia is under sanctions that drown out any commodity signal. Uzbekistan and the large West African producers run pegged or managed exchange rates, and a pegged currency cannot transmit a gold signal at all. Peru and Ghana are workable candidates and are flagged as natural extensions.

1.2 Why everything is measured in yen

Because gold is priced in dollars, testing AUD/USD against gold-in-USD puts the dollar inside both series. When the dollar weakens, gold rises and AUD/USD rises at the same time, even if Australia's gold exports did nothing. The test would find a gold link that is really a dollar link. Section 2 shows exactly this happening to the rand.

A clean test needs a neutral measuring stick: a currency that is not the dollar, floats freely with deep markets, and belongs to a country with no meaningful gold production of its own. The Japanese yen fits all three conditions. Every price in this report, including gold, is therefore measured in yen, and the backtest trades yen crosses such as AUD/JPY. Results are cross-checked in dollar terms to show exactly how much the measuring stick changes the answer. One honest note: profits earned in yen crosses carry some exposure to the yen's own swings, so re-running the whole pipeline with the euro is a natural robustness check.

2. The Dollar Effect

Exhibit 1 is the single most important chart in this report. In dollar terms the rand looks strongly related to gold, though with a negative sign (b -0.57, R2 0.53). Re-priced in yen, the relationship mostly dissolves into two unrelated trends: a rising gold price and a rand that has been depreciating for twenty years.

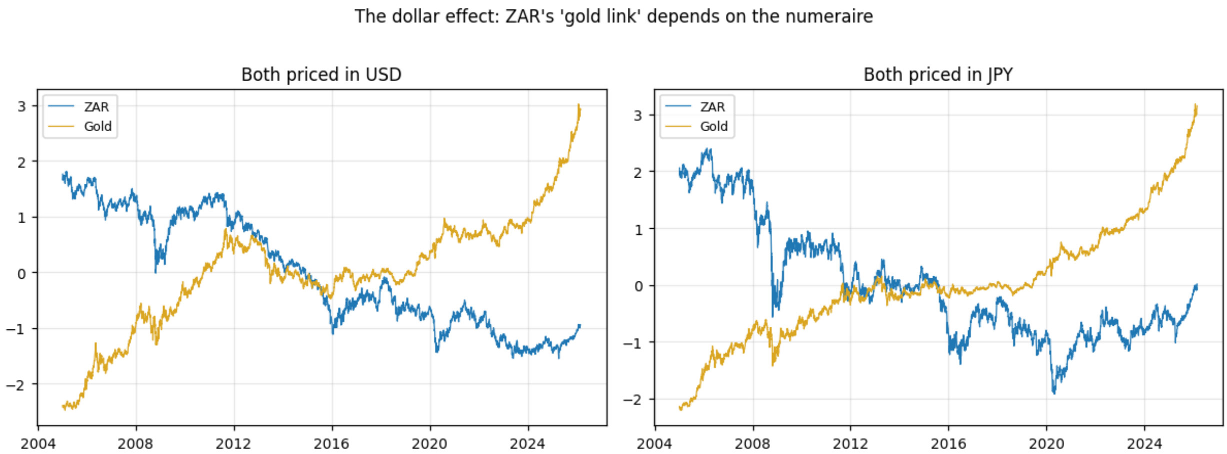

Exhibit 1: ZAR vs. gold, both priced in USD (left) and JPY (right). Normalized log levels.

Any analysis run purely in dollar terms inherits this distortion. It is probably the most common mistake in published gold-currency commentary.

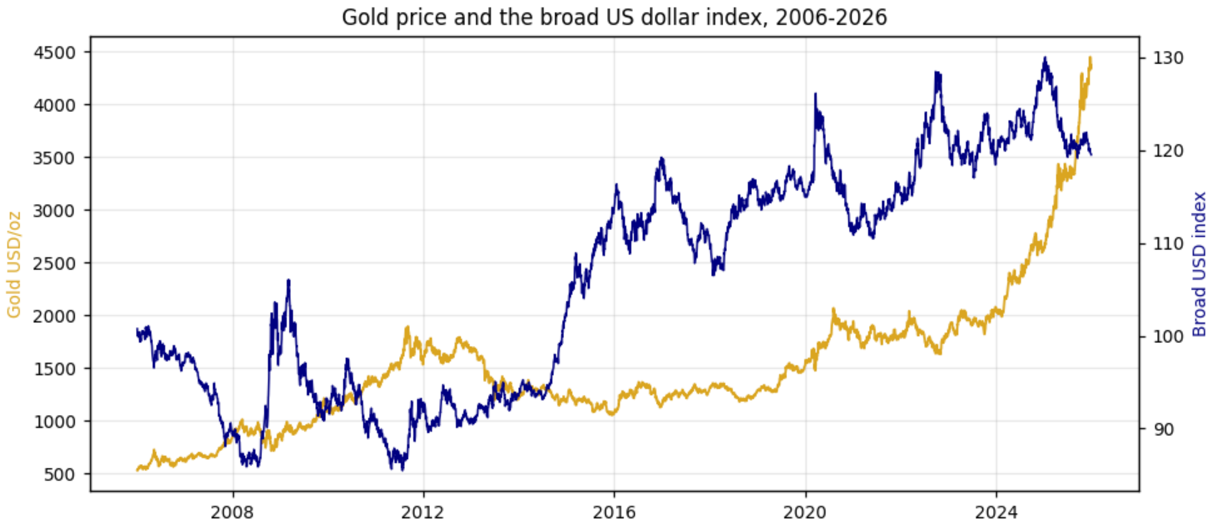

Exhibit 2: Gold (USD/oz) vs. the broad USD index, 2006-2026. The index begins in January 2006, which is why the chart starts there. This inverse relationship sits underneath every dollar-denominated gold chart.

3. Australian Dollar: A Genuine Link, But Only Recently

The full-sample regression is unimpressive (b 0.07, R2 0.11). Splitting the sample into periods changes the story completely:

Period	Gold beta	Newey-West t	R2
2005-2012	-0.02	-1.3	0.00
2013-2019	-0.51	-3.2	0.11
2020-2026	+0.23	11.0	0.63

The Australian dollar became a gold currency in the 2020s, the decade in which gold ran from $1,500 to above $5,000 and central banks bought record amounts. The critical test is whether the link survives controls, and it does: with the broad dollar and the VIX included in the regression, gold keeps a coefficient of 0.16 with a t-statistic of 12.2. For AUD, gold is not just the dollar in disguise.

One caveat belongs next to the 63% figure. It comes from a regression on trending price levels, and Section 6 shows there is no long-run anchor between the series. So read the 63% as a description of how tightly the two prices moved in that era, not as proof of a permanent bond. The strong t-statistics, and the fact that the result repeats across different tests, are what carry the conclusion.

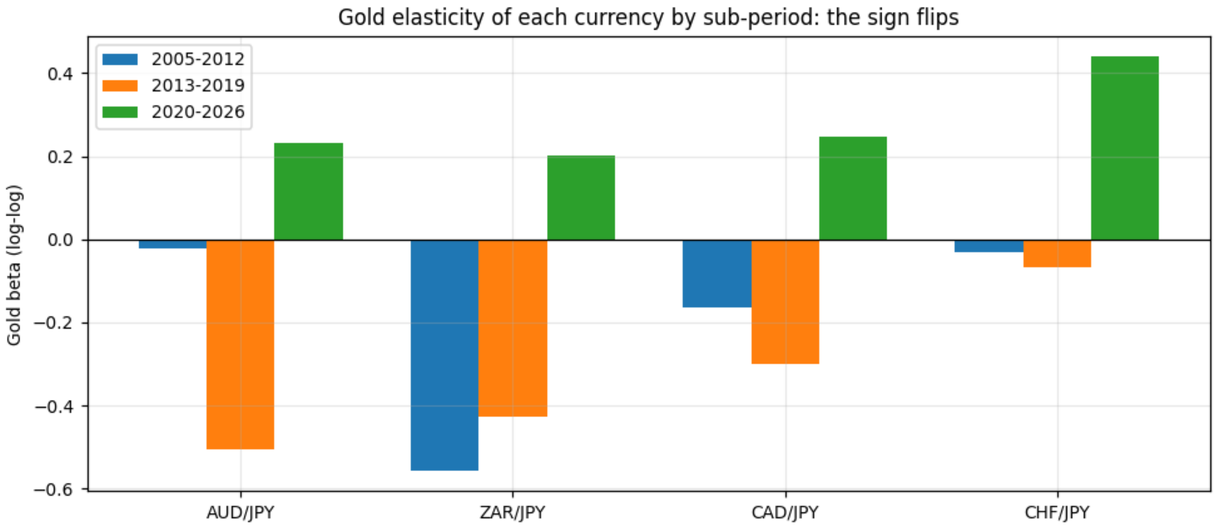

Exhibit 3: Gold sensitivity by sub-period. Negative or zero before 2020, positive everywhere after. The sign flips.

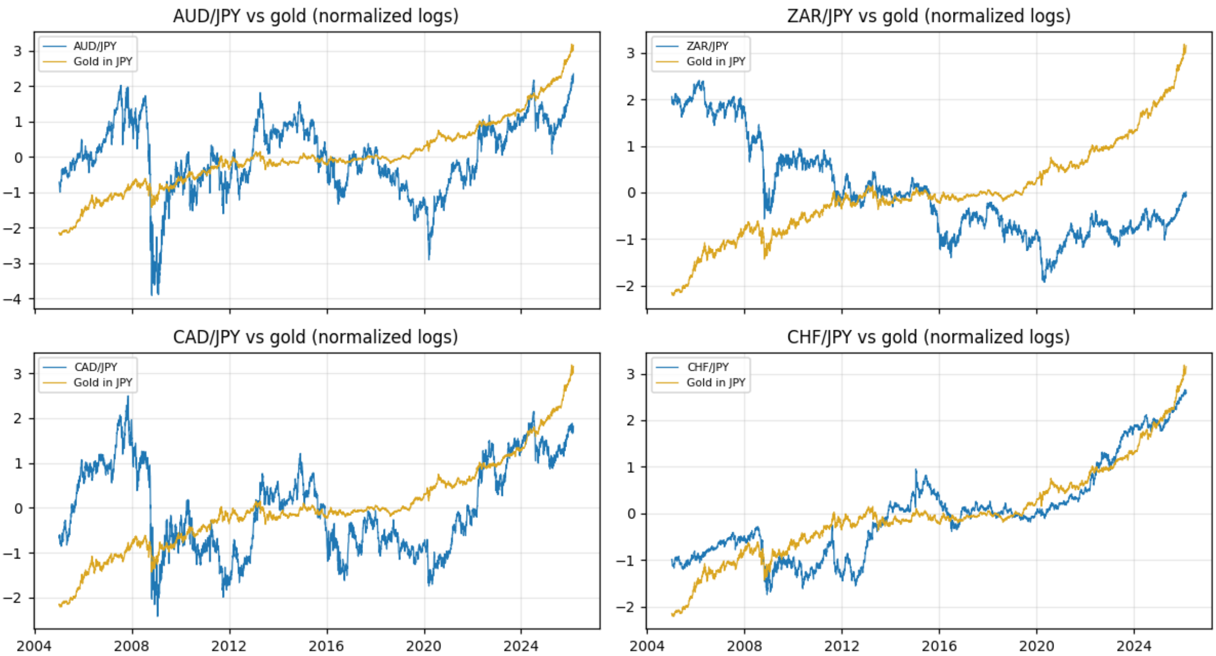

Exhibit 4: Each currency vs. gold in JPY terms, normalized log levels, 2005-2026.

4. South African Rand: Title Retired

The full sample in yen terms gives a beta of -0.37 (R2 0.50), which is the wrong sign for a gold currency. With controls, gold stays negative (-0.15) while the broad dollar absorbs the entire relationship (b -1.22). In plain terms: the rand follows the dollar and global risk appetite, not gold. The reason is structural. South African gold output has been declining for two decades, from world number one to outside the top five, while platinum metals, coal and, above all, global market mood now dominate the currency. The positive 2020-26 numbers (b 0.20) reflect the global regime rather than the mining sector, and the controls model makes that explicit.

5. Canadian Dollar and Swiss Franc: Control and Cautionary Tale

CAD (control). Gold effects are small in the full sample (R2 0.09). With controls, though, the coefficient is positive and significant (b 0.11, t 5.2), and the 2020-26 R2 reaches 0.60. The Canadian dollar has actually been a better gold currency than the rand this decade, which was not the expected result.

CHF (cautionary tale). The franc posts the highest gold fit of all four currencies (full-sample R2 0.78, t of 24 with controls) despite having no domestic mining at all. Gold and the franc are the two classic safe-haven assets, so they trend together against everything else. This is the clearest possible proof that a big gold correlation does not require a single gram of gold exports, and it is exactly why the exporter regressions in this report needed controls.

6. The Statistical Evidence, Translated

Do these prices wander without an anchor? Yes. All the price series drift like a walker with no home (they are I(1) in technical terms), which is exactly why naive correlations of raw prices mislead.

Is there a rubber band pulling gold and each currency back together? No. This is what cointegration tests measure, and they reject it everywhere: Engle-Granger fails for all four currencies (AUD comes closest at p = 0.07; ZAR is nowhere at p = 0.88), and the Johansen test agrees in every case. Where an adjustment speed can be estimated at all, closing half of a gap would take 190 to 350 trading days. In practice, when gold and a currency drift apart, nothing dependable pulls them back.

Who moves first? This is what Granger causality tests ask: do yesterday's gold moves predict today's currency moves, or the reverse? The answers split the four currencies into two neat camps:

Direction	AUD	ZAR	CAD	CHF
Gold leads the currency (p-value)	0.001	0.39	0.011	0.39
The currency leads gold (p-value)	0.06	0.004	0.27	0.007

Gold moves first for the producer currencies: news lands in the gold market and seeps into the currencies of the countries that mine it. For the rand and the franc, information flows the other way, because these currencies are barometers of global mood, and the mood moves gold. The direction of the arrow itself tells you what kind of currency you are looking at.

Can gold explain currency moves on fresh data? Modestly, yes. Using the same day's gold move, a gold-based model beat the humble 'tomorrow equals today' benchmark by 9 to 12% for all four currencies. But note the fine print: that test uses the same day's gold price, so it explains rather than predicts.

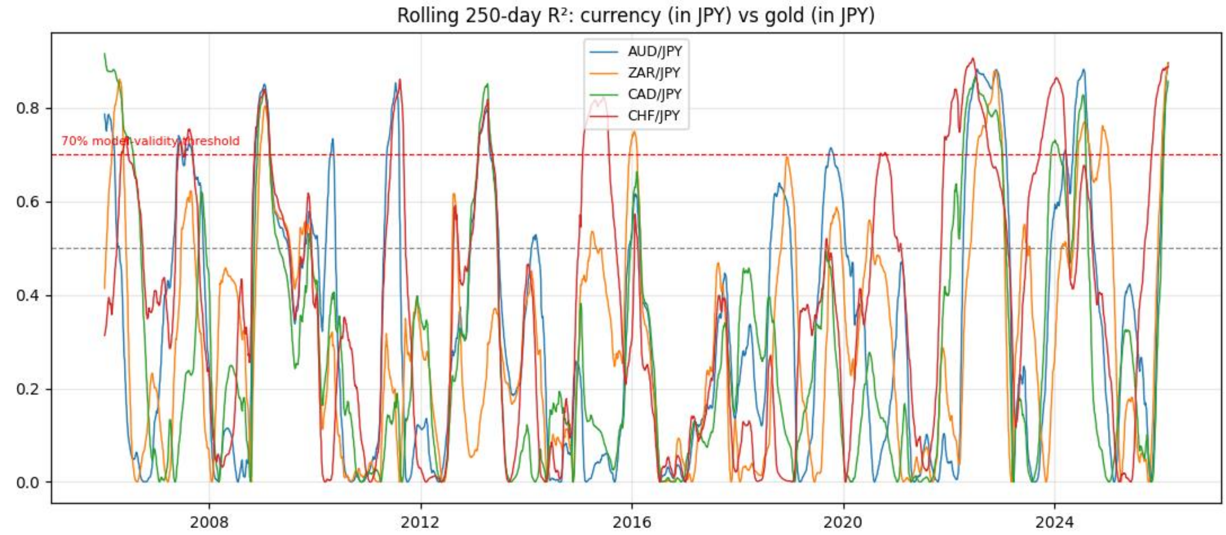

Exhibit 5: Rolling one-year fit (R2) with gold, for each currency. The link comes in episodes, above the 70% line only 7 to 16% of the time, clustered in 2008-09, 2011-13 and 2022-26. The rolling window needs a year of data, hence the 2006 start.

Exhibit 5 may be the most useful chart in the report. Every currency's gold link spends most of its life below 50% and jumps above 70% only in bursts. Being a gold currency is something a currency does occasionally, not something it is.

7. Strategy Backtest: Can You Make Money From This?

Currency forecasting is a humbling sport, so the test here is deliberately simple. Fit a rolling 50-day model of the currency against gold. Only act when the model currently fits well (rolling R2 of at least 50%). When the currency breaks two standard deviations out of its band, follow the move. Exit after ten days or at a 2% stop. The backtest runs from August 2014 (after a 50-day warm-up on data starting June 2014) through February 2026, before costs:

Currency	Trades	Total return	Win rate	Assessment
AUD	43	+4.5%	53%	Marginally positive; peaks at +9.2% with tuned settings, but nearby settings lose money
ZAR	45	-29.0%	36%	Uninvestable
CAD	40	-5.0%	53%	Negative
CHF	45	-8.9%	44%	Negative

Only the Australian dollar finishes above water, and thinly. Three qualifications keep this result honest. First, the +9.2% figure is in-sample optimization: it is the best cell of a 16-setting grid searched over the whole period, not a result the strategy would have found in advance. Second, at roughly ten basis points of average profit per trade, realistic spreads and slippage in AUD/JPY would eat much or all of the edge. Third, the returns are sums of per-trade results on unit positions rather than a compounded portfolio return, and no Sharpe ratio or drawdown figures are claimed.

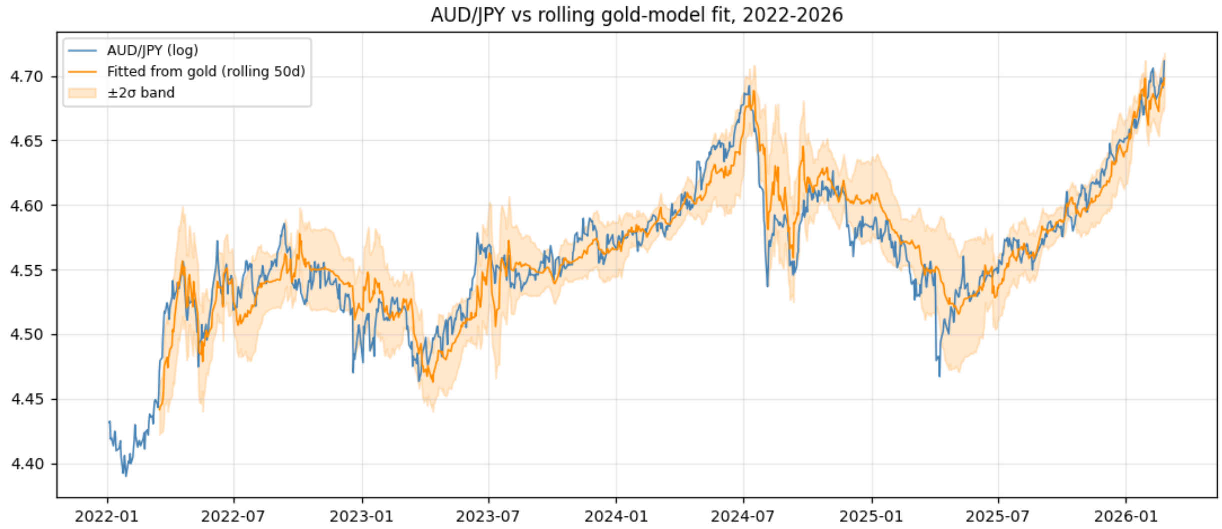

Exhibit 6: AUD/JPY vs. the rolling gold-model fit with 2-sigma bands, 2022-2026. The recent window is shown so the bands are readable; the model runs on the full history.

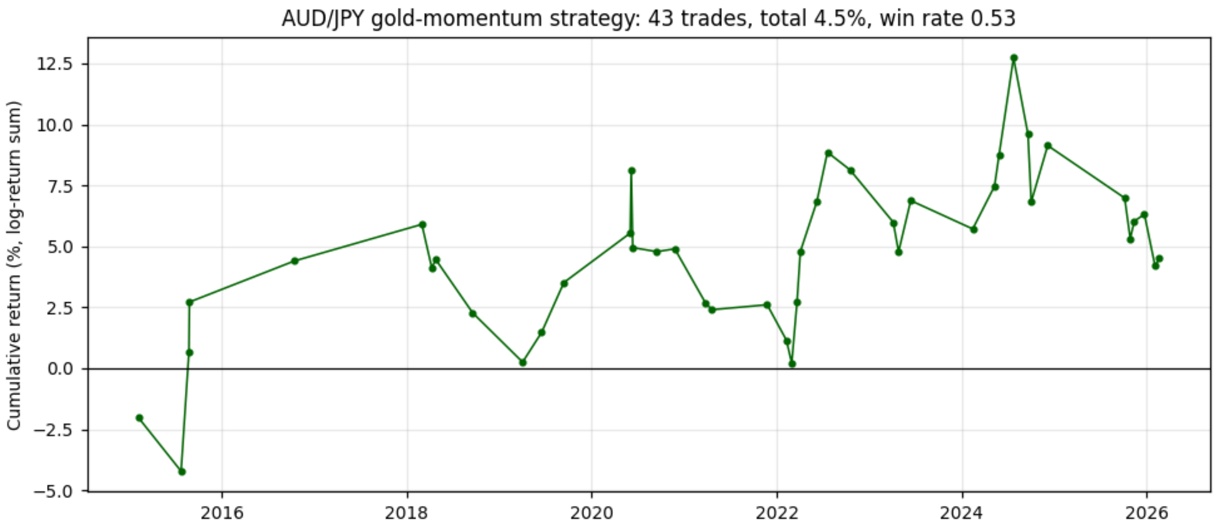

Exhibit 7: AUD strategy equity curve. 43 trades, +4.5% cumulative. The x-axis points are trade exit dates.

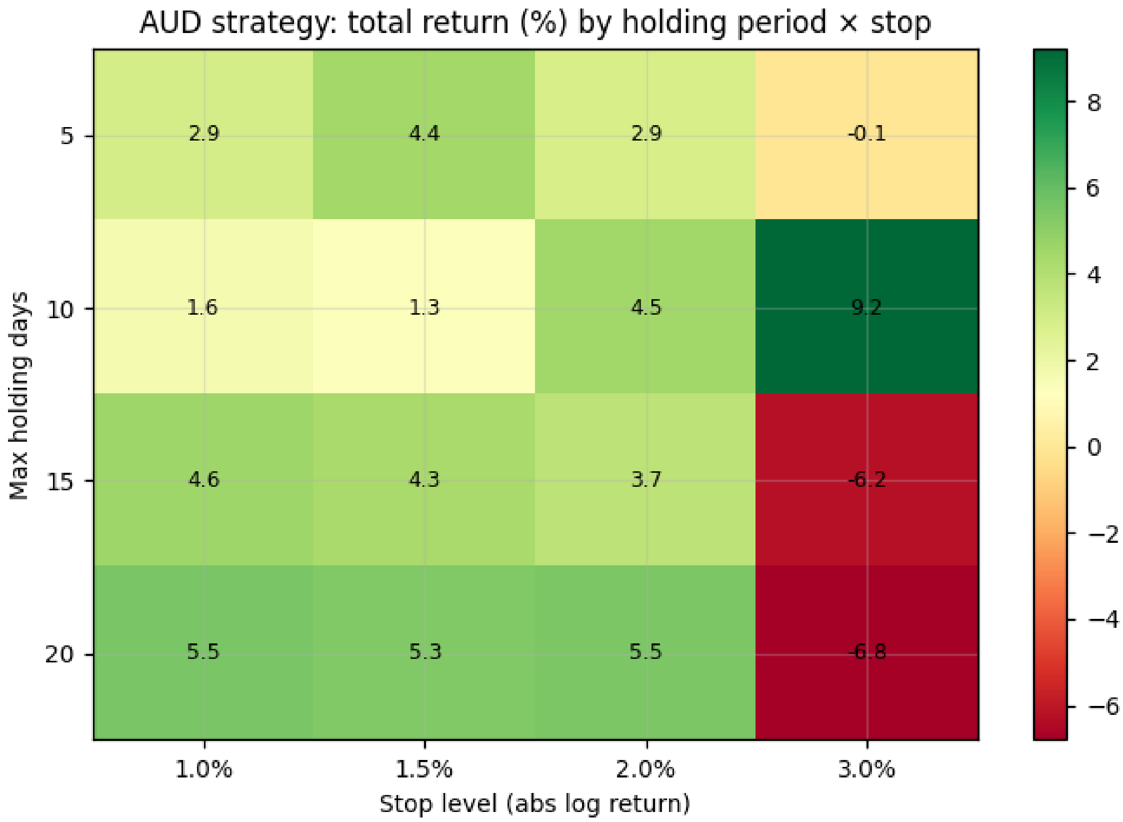

Exhibit 8: AUD total return across the holding-period and stop-level grid. Adjacent cells flip sign, which is the signature of a fragile edge.

The lesson is not that markets are unbeatable. It is that the gold-currency relationship flips regimes too often for a fixed rule. If a tradable version exists, it must switch itself off outside the high-fit episodes of Exhibit 5, and even then expectations should stay modest.

8. What This Means

For investors. Do not read a gold chart as a currency forecast. The link switches on in weak-dollar, high-fear eras and off otherwise. Watch the dollar and US real interest rates first, because they sit behind both sides of the chart. If you trade the link at all, trade it only when a rolling fit confirms it currently exists, and size the position for a small edge.

For policymakers in gold-exporting countries. Gold booms genuinely can strengthen the currency when the global weather cooperates, as Australia after 2020 shows. But there is no long-run anchor, and South Africa shows how completely gold-currency status can fade as mines decline. A gold windfall deserves the same caution as any commodity windfall: save some of it, hedge some of it, and do not build the budget on it.

For researchers. The who-moves-first test cleanly separated producer currencies from risk currencies in this sample, which makes an inexpensive, repeatable classification rule. Natural next steps: the Peruvian sol, the Ghanaian cedi and the West African producers, a daily real-interest-rate control, and a euro-numeraire robustness run.

9. Conclusion

Is there such a thing as a gold currency? Sometimes, somewhere, partly. The Australian dollar has earned the label since 2020 and keeps it even after the dollar and fear are accounted for, which is the strongest single finding in this report. The South African rand, the currency that made the phrase famous, no longer qualifies: measured honestly, its gold link is the dollar effect plus a long depreciation. The Swiss franc imitates a gold currency through the safe-haven door without a single mine, which is the clearest possible warning against reading correlation as geology. And nothing ties any of these currencies to gold in the long run: the rubber band the tests searched for simply is not there.

The gold price is a mirror of the world's fears and its interest rates. The currencies of gold-mining countries are reflections in that mirror: real, but visible only when the light is right. Every confident claim that gold strength must lift a mining currency deserves the question this report kept asking: measured in what, controlling for what, and in which era? The answer, more often than not, changes the story.

Risk Factors and Limitations

- **Real rates.** No daily real-rate (TIPS) data was available in the free sources used, so real interest rates are discussed qualitatively rather than included in the regressions.
- **Data coverage.** The broad dollar index covers 2006 to December 2025; the daily gold fix in the mirror used ends February 2026.
- **Universe.** Smaller gold exporters (Peru, Ghana and others) are not yet covered.
- **Costs.** Backtest results are before transaction costs, slippage and funding.
- **Regime dependence.** Recent results are dominated by the 2020-26 era, which may not persist. Regimes can only be identified after the fact.

Appendix A: Data

Series	Source (mirror)	Range used
Gold PM fix, USD (daily)	LBMA via forex-centuries	2005 to 2026-02
Fed H.10 daily FX (local-per-USD)	datasets/exchange-rates	2005 to 2026-02
Broad USD index (DTWEXBGS)	FRED via forex-centuries	2006 to 2025-12
VIX daily close	CBOE via datasets/finance-vix	2005 to 2026-02
Gold monthly average	LBMA via datasets/gold-prices	context only

Crosses are built as yen per unit of currency (AUDJPY = (JPY/USD)/(AUD/USD)); gold in yen is the dollar fix multiplied by JPY/USD. All models use natural logs. The full daily panel, every model output and the backtest ship in the companion workbook Gold_Money_Data_v2.xlsx and the repository (github.com/maitrayee196/Gold-Prices-vs-Exchange-Rates).

Appendix B: Selected References

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